Worksheet 4

Bivariate distributions and linear fitting – EXERCISES

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Exercise 1 – Correlation and least-squares line

FERSET observes annual customer numbers x (hundreds) and revenue y (in 10^4 euros):

Year	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
x (hundreds of customers)	5	6	7	9	8	11	12	13	14	16
y (10^4 euros)	9	10	14	16	19	20	23	26	29	32

1. Build the table $x_i, y_i, x_i^2, y_i^2, x_i y_i$ and calculate the sums.
2. Calculate $\bar{x}$, $\bar{y}$, $V(x)$, $V(y)$, σ_x, σ_y, $Cov(x, y)$, and R.
3. Interpret R using the rule stated in the course.
4. Find the least-squares line in the course convention: $\hat{y} = ax + b$, where $a = Cov(x, y)/V(x)$.
5. Forecast revenue if the number of customers is 3000.

Exercise 2 – Customer numbers as a function of time

Set $t = 1$ for 2005, ..., $t = 10$ for 2014. Calculate $R(t, x)$, find $\hat{x} = at + b$, forecast the number of customers in 2016 ($t = 12$), then use Exercise 1 to forecast revenue. **Exercise 3 – Mayer line**

For points $(1, 7), (2, 10), (3, 10), (4, 14), (5, 14)$, form the two sub-clouds used in the course, calculate their mean points A and B, and find the Mayer line.